Lecture 26

Evolution of the scattering data

KdV moves the potential while its scattering data obey simple time laws. We prove the bound-state law under an isolated-branch hypothesis. The normalization at spatial infinity requires a separate estimate.

Objectives. *We prove constancy of an isolated negative eigenvalue and evolution of its right norming constant. We solve the rank-one inverse equation and recover the travelling pulse. The reflection phase is stated with a derivation sketch.*

By the end of this lecture you will be able to:

- Justify the bound-state commutator pairing on its actual domain.
- Derive the time law for the right-normalized norming constant.
- Solve the one-state Marchenko equation and locate the negative-data pole.

26.1 The isolated branch hypothesis

Use the real solution $u \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}))$ of KdV from Lecture 24, equation (24.5). Recall

$$L = -\partial_x^2 + u, \qquad B = -4\partial_x^3 + 6u\partial_x + 3u_x, \qquad L_t = [B, L].$$

For each fixed time, L is self-adjoint on $H^2(\mathbb{R}, \mathbb{C})$: the free operator has this domain and real bounded multiplication is a bounded self-adjoint perturbation. Here the L^2 inner product is antilinear in its first argument. This uses Kato, V, Theorem 4.3, p. 287 (Kato–Rellich): bounded real multiplication has relative bound zero with respect to the free operator.

Definition 26.1 (Differentiable isolated bound-state branch)**.** On an open interval I, a *differentiable isolated branch* consists of $\lambda \in C^1(I; (-\infty, 0))$ and a nonzero real $g \in C^1(I; H^2(\mathbb{R}))$ such that

$$L(t)g(t) = \lambda(t)g(t).$$

At every $t \in I$, require $g(t) \neq 0$ and let $\lambda(t)$ be an isolated simple eigenvalue of $L(t)$. Simple means that its eigenspace has dimension one.

This is an additional hypothesis on the branch, rather than a consequence of the differential identity alone. The local perturbation input is Kato, *Perturbation Theory for Linear Operators*, second edition, IV, Theorem 3.16, pp. 212–213, and supplementary note II.3, pp. 568–569: an isolated finite spectral subspace persists locally, and a simple eigenpair has the parameter regularity of the reduced operator. For the present common domain, the resolvent identity differentiates because $u(t)$ is C^1 in the bounded-multiplication norm. The contour projection is therefore C^1, also into H^2 by the graph norm, and its one-dimensional range supplies the local branch. These are *cited perturbation results*; they do not assert a global evolution operator for B. An original scan is available at Kato, original text. We prove the evolution on any interval carrying the stated branch.

26.2 Constancy of the eigenvalue

The eigenfunction has all Sobolev derivatives. Indeed, $g_{xx} = (u-\lambda)g$ and bounded derivatives of u successively give $g \in H^m$ for every integer m. The same equation and its time derivative bootstrap the assumed $C^1(H^2)$ dependence to $C^1(H^m)$ on compact time intervals. In one dimension, H^1 functions have continuous representatives tending to zero at both ends. This applies to every derivative of g.

Differentiate the eigenvalue equation and pair with g. Self-adjointness cancels the g_t terms, leaving

$$\lambda' \|g\|_2^2 = (g, u_t g) = (g, [B, L]g) = \lambda(g, Bg) - (Lg, Bg) = 0.$$

The Lax identity here is the pointwise differential formula proved in Lecture 24, now applied to the smooth g. Both compositions belong to L^2: in particular $Bg \in H^2$ because $g \in H^5$ and the coefficients have bounded derivatives. Thus moving L across the inner product is legitimate. It follows that $\lambda = -\kappa^2$ for a constant $\kappa > 0$ on I.

26.3 The right normalization at negative energy

Lecture 25 constructs Jost solutions at real nonzero k. For the negative energy just obtained, we give the required extension. Write $f(x,t) = e^{-\kappa x} m(x,t)$ on a right half-line. The right normalization is $m(x,t) \to 1$ as $x \to +\infty$. The corresponding Volterra equation is

$$m(x,t) = 1 + \int_x^\infty h_\kappa(y-x) u(y,t) m(y,t)\, \mathrm{d}y, \qquad h_\kappa(s) = \frac{1 - e^{-2\kappa s}}{2\kappa}. \tag{26.1}$$

Its kernel satisfies $0 \le h_\kappa(s) \le (2\kappa)^{-1}$ for $s \ge 0$. Successive integration on ordered simplices bounds its nth iterate by $U(x)^n/n!$, where $U(x) = (2\kappa)^{-1} \int_x^\infty |u(y,t)| \mathrm{d}y$. The series converges uniformly on each right half-line. It gives a unique bounded solution there, with $|m| \le e^{U(x)}$ and $|m-1| \le e^{U(x)} - 1$. Differentiating in x proves $f_{xx} = (u + \kappa^2) f$. Ordinary differential-equation uniqueness extends f to the whole line.

The estimates can be uniform for t in any compact $J \subset I$. Set

$$U_J(x) = \frac{1}{2\kappa} \int_x^\infty \sup_{t \in J} |u(y,t)| \mathrm{d}y, \qquad V_J(x) = \frac{1}{2\kappa} \int_x^\infty \sup_{t \in J} |u_t(y,t)| \mathrm{d}y.$$

Both tend to zero by the uniform Schwartz bounds. Termwise time differentiation of the Volterra series is valid: the differentiated terms have the convergent majorant $V_J(R) e^{U_J(R)}$ on $x \ge R$. Equivalently $m_t = T m_t + T_t m$, where T is the integral operator in (26.1). Iterating this equation gives

$$\sup_{t \in J} |m_t(x,t)| \le e^{2U_J(x)} V_J(x) \longrightarrow 0.$$

The spatial derivatives needed in B also have controlled limits. Differentiation of the bounded kernel gives

$$\begin{aligned} m_x &= -\int_x^\infty e^{-2\kappa(y-x)} u(y,t) m(y,t) \mathrm{d}y, \\ m_{xx} &= um + 2\kappa m_x, \\ m_{xxx} &= u_x m + u m_x + 2\kappa m_{xx}. \end{aligned}$$

Each tends to zero uniformly on J as $x \to +\infty$. Consequently $e^{\kappa x} B f \to 4\kappa^3$ and $e^{\kappa x} f_t = m_t \to 0$. These conclusions use the Volterra estimate, not merely $C^1(L^2)$ dependence of an eigenfunction.

The constructed f is the right-normalized bound state. To see this, fix a compact time interval $J \subset I$ and choose R with $U_J(R) < \log 2$. Then $|m-1| \le e^{U_J(R)} - 1 < 1$ on $[R,\infty)\times J$, so $f(R,t) \neq 0$ for every $t \in J$. A second independent solution there is $f(x)\int_R^x f(y)^{-2}\mathrm{d}y$, which grows like $e^{\kappa x}$. Since $g \in L^2$, its growing coefficient must vanish. Thus g is a nonzero multiple of f. The ratio at the fixed point R is C^1 in time. It follows that f is a real $C^1(H^m)$ branch for every m. This also proves that the right normalization fixes its sign.

Definition 26.2 (Right norming constant)**.** As in Lecture 25, equation (25.6), the *right norming constant* is

$$c(t) = \left(\int_{\mathbb{R}} f(x,t)^2 \mathrm{d}x\right)^{-1} > 0. \tag{26.2}$$

26.4 Evolution of the bound-state data

Theorem 26.3 (Bound-state evolution)**.** *Under the standing Schwartz solution and isolated-branch hypotheses, κ is constant. The right-normalized bound state and its norming constant satisfy*

$$f_t = (B - 4\kappa^3)f, \qquad c(t) = c(t_0)e^{8\kappa^3(t-t_0)} \quad (t, t_0 \in I).$$

Proof. Constancy of κ was proved above. Differentiate $(L+\kappa^2)f = 0$ and use $L_t = [B,L]$. The result is

$$(L+\kappa^2)(f_t - Bf) = 0.$$

Both terms are in H^2, by the proved $C^1(H^m)$ regularity. Simplicity makes $f_t - Bf = \gamma(t)f$. Multiplying by $e^{\kappa x}$ and taking the proved right limits gives $\gamma = -4\kappa^3$.

Let $N = \int f^2$. All derivatives of f tend to zero at the endpoints. Integration by parts gives

$$\int_{\mathbb{R}} fBf\,\mathrm{d}x = \left[-4\left(ff'' - \frac{1}{2}(f')^2\right) + 3uf^2\right]_{-\infty}^{+\infty} = 0.$$

The $C^1(L^2)$ chain rule now yields $N' = 2\int ff_t = -8\kappa^3 N$. Since $c = N^{-1}$, we obtain $c' = 8\kappa^3 c$ and the displayed time law. ∎

The reflection coefficient has the corresponding law $r(k,t) = r(k,t_0)e^{8\mathrm{i}k^3(t-t_0)}$ for real $k \neq 0$. Its derivation is *sketched* here. With the scattering basis of (25.1), comparison at the two infinities gives

$$(f_+)_t = Bf_+ - 4\mathrm{i}k^3 f_+, \qquad (f_-)_t = Bf_- + 4\mathrm{i}k^3 f_-.$$

The signs follow from $-4\partial_x^3 e^{\pm\mathrm{i}kx} = \pm 4\mathrm{i}k^3 e^{\pm\mathrm{i}kx}$. Differentiate $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. Independence of the right basis then gives $a_t = 0$ and $b_t = 8\mathrm{i}k^3 b$. The omitted step is justification of time differentiation and asymptotic comparison for the real-energy Jost functions; Sheet 26 supplies those estimates. It is separate from the proved negative-energy argument.

26.5 One-state reconstruction

We use the *cited right Marchenko inverse theorem* for real rapidly decaying potentials: the reflection coefficient, negative energies and positive right norming constants determine the potential through

$$K(x,y)+F(x+y)+\int_x^\infty K(x,z)F(z+y)\mathrm{d}z=0,\qquad y\geq x,$$
$$F(s)=\sum_j c_j e^{-\kappa_j s}+\frac{1}{2\pi}\int_{\mathbb{R}} r(k)e^{\mathrm{i}ks}\mathrm{d}k,\qquad u(x)=-2\frac{\mathrm{d}}{\mathrm{d}x}K(x,x).$$

The derivative in the reconstruction is the total derivative along the diagonal. The precise cited input is the zero-background specialization of Egorova–Grunert–Teschl, Theorems 4.3–4.4, author preprint, Section 4, recalled in Lecture 25. Their $(\gamma_j^+)^2$ is our c_j. See Dunajski, *Solitons, Instantons, and Twistors*, §2.2.3, equations (2.2.8)–(2.2.10), pp. 30–31; the rank-one calculation is §2.3.1, p. 34. We use the inverse theorem as an input, rather than claiming its proof from the real-energy construction.

Example 26.4 (A single positive norming constant)**.** Take reflectionless data $r=0$, one $\kappa>0$ and $c>0$. Then $F(s)=ce^{-\kappa s}$. The integral equation forces $K(x,y)$ to be proportional to $e^{-\kappa y}$. Writing $K(x,y)=\zeta_M(x)e^{-\kappa y}$ and evaluating the integral gives

$$\zeta_M+ce^{-\kappa x}+\frac{c}{2\kappa}e^{-2\kappa x}\zeta_M=0,\qquad K(x,y)=-\frac{ce^{-\kappa(x+y)}}{1+\frac{c}{2\kappa}e^{-2\kappa x}}.$$

The denominator is strictly positive. Differentiation along $y=x$ gives

$$u=-\frac{4\kappa ce^{-2\kappa x}}{\left(1+\frac{c}{2\kappa}e^{-2\kappa x}\right)^2}=-2\kappa^2\operatorname{sech}^2\bigl(\kappa(x-x_c)\bigr),\qquad x_c=\frac{1}{2\kappa}\log\frac{c}{2\kappa}.$$

Prescribe $c(t)=c(t_0)e^{8\kappa^3(t-t_0)}$ and reconstruct these one-state reflectionless data at each time. The centre then satisfies $x_c(t)=x_c(t_0)+4\kappa^2(t-t_0)$. The resulting family solves KdV by the direct substitution verified in Lecture 23. Thus the explicit solution realizes the proved norming law; this calculation needs no general theorem that the KdV flow preserves the number of bound states.

For this potential the right-normalized bound state is $f=(e^{-\kappa x_c}/2)\operatorname{sech}(\kappa(x-x_c))$. Its right tail is $e^{-\kappa x}$, and $\int f^2=e^{-2\kappa x_c}/(2\kappa)$. Thus its norming constant is indeed $2\kappa e^{2\kappa x_c}=c$. Translation changes c while leaving the eigenvalue and zero reflection unchanged. The norming constant therefore contains location information.

A negative value of c cannot represent (26.2). It also makes the algebraic reconstruction singular: the denominator vanishes at $x_*=(2\kappa)^{-1}\log(-c/(2\kappa))$. Its derivative there is $2\kappa\neq 0$, while the numerator is nonzero. The kernel has an actual pole, so the reconstructed potential is singular.

Summary. For a differentiable isolated negative branch of a Schwartz KdV solution, the eigenvalue $-\kappa^2$ is constant. A right-tail Volterra estimate fixes the temporal normalization: $f_t=(B-4\kappa^3)f$ and $c'=8\kappa^3 c$. The real-energy reflection phase was sketched with its analytic step identified. Conditional on the right inverse theorem, one positive norming constant reconstructs the negative pulse moving at speed $4\kappa^2$. A negative norming coefficient produces a pole.

Self-check. Why does the commutator pairing vanish? Which domain condition is needed?

Answer. Both terms reduce to $\lambda(g, Bg)$ by $Lg = \lambda g$ and self-adjointness. Moving L requires $Bg \in H^2$, supplied by the eigenfunction's H^5 regularity. A formal test-domain identity alone would not supply this step.

Self-check. Why does the right-normalized state satisfy $f_t = Bf - 4\kappa^3 f$ rather than $f_t = Bf$?

Answer. The leading right tail of Bf is $4\kappa^3 e^{-\kappa x}$. The Volterra estimate gives $e^{\kappa x} f_t \to 0$. The subtracting scalar term preserves the fixed unit right-tail coefficient.

Self-check. If $\kappa = 1$ and $c(t_0) = 2$, find the pulse centre and its motion.

Answer. The centre is zero at t_0. Since $c(t) = 2e^{8(t-t_0)}$, the centre is $x_c(t) = 4(t - t_0)$. The pulse minimum remains -2.

Self-check. At $\kappa = 1$ and $c = -2$, where does the Marchenko denominator vanish? Can these data describe a smooth bound-state potential in the stated class?

Answer. The denominator is $1 - e^{-2x}$ and vanishes at zero. The kernel numerator there is nonzero, so the reconstruction has a pole. The data also contradict $c = (\int f^2)^{-1} > 0$.